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THE POSTMAXIMUM SPECTRUM OF χ CYGNI*

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ABSTRACT

The spectrum of the long-period variable χ Cygni combines characteristics of classes M and Numerous bright lines are unusually intense. Spectrograms extending from the blue into the ultraviol dispersion 10 A/mm, were obtained over the interval from 41 to 162 days after maximum; the cor sponding range of magnitude was from 6.3 to 11.4.

Dark lines.—The atomic-line spectrum resembles that of R Andromedae, class Se, but the *TiO* ba are as strong as in class M6. The velocities exhibit a very slight increase (algebraic) with advancing pha Differences between various cycles are small. Residuals of individual lines behave as in other red va ables.

Bright lines.—After maximum, bright lines of *Fe* I increased in number from 30 to almost 100. M of them grew stronger throughout the period of observation, but a few high-excitation multiplets l their maximum intensities about 114 days after maximum. The behavior of multiplet (2) illustrates tendency of bright lines to escape from the effects of the reversing layer as the phase advances. A culiarity of physical interest is the inordinately high intensity of bright lines from upper levels z and z^5G_2. A third level possibly emphasized is w^5D_4, and others not so outstanding are y^5F_1 and y^5F_2. *Zr* I level x^1G_4 is the only one that gives use to bright lines. In several other elements possible examp of excitation by line coincidence are noted.

Numerous bright lines of *Co* I and *Ni* I have been identified. Multiplets having certain *lower* levels emphasized. This fact probably finds its explanation in the different relative intensities of absorpt and emission lines at various excitation potentials.

Numerous bright lines of *Fe* II became outstanding with advancing phase; forbidden lines appea about 100 days after maximum and strengthened rapidly during the next two months. Other bright li discussed are those of *H*, *Mg* I, *Si* I, *Mn* I, *In* I, and the curious displaced components of *Al* I, *Ca* and other elements.

Velocities from lines of *H*, *Fe* II, and *Ti* II agree with one another and vary little with phase; veloci from *Fe* I lines increase slightly with excitation potential and with phase. Observations of R Androme and χ Cygni suggest that Se variables have smaller periodic variations in the positions of lines b bright and dark than do Me variables.

Wave lengths of numerous bright lines measured on Plate Ce 3505, phase +162 days, magnit 11.4, are collected in Table 13. The table indicates how the intensity behaves as the phase advances number of strong lines remain unidentified.

The variability of χ Cygni, 194632, was discovered in 1686 by G. Kirch as a res of comparing the sky with Bayer's star atlas of 1603. The only variable stars discove earlier are *o* Ceti (1596) and β Persei (1669). The period of χ Cygni is 406.5 days wit mean magnitude range of from 5.3 to 13.3. The extreme range has probably been mag.; thus at the brighter maxima the star is about ten thousand times as bright as the fainter minima.

The spectrum was classified Md at Harvard. Later observations with slit spect graphs have shown that it combines characteristics of classes M and S. Numer bright lines are unusually intense. Previous observations of χ Cygni with slit spect graphs are listed in Table 1.

In the present investigation, spectrograms having dispersion 10.3 A/mm were tained with the 100-inch telescope at four epochs after maximum light in 1944 and one epoch in 1945 (Table 2). All were on the II*a*-O emulsion supplied by the Resea Laboratory of the Eastman Kodak Company. In 1944 the plates were baked at 50 for 3 days prior to exposure.

The dates of observation are marked on the light-curve in Figure 1. I am again debted to Mr. Leon Campbell of the Harvard Observatory for the photometric da

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TABLE 1

PREVIOUS OBSERVATIONS WITH SLIT SPECTROGRAPHS

Observatory	Observer	Year	Reference
Potsdam.....	G. Eberhard	1901, 1902	*Ap. J.*, **18**, 198, 1903
Univ. Michigan...	P. W. Merrill	1914	*Pub. Obs. Univ. Michigan*, **2**, 45, 1916
Victoria.....	F. S. Hogg	1932	Unpublished
Mount Wilson....	P. W. Merrill	1920 1928 1921 1932 1923 1937 1926 1938 1927 1939	*Mt. W. Contr.*, *Ap. J.* No. 200 **53**, 185, 1921 264 **58**, 215, 1923 399 **71**, 285, 1930 539 **83**, 272, 1936 642 **93**, 40, 1941 649 **94**, 171, 1941

TABLE 2

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Plate Ce	Date	JD 243	Vis. Mag.	Phase (Days)	Exp. Min.	Rad Vel. from Abs. Lines	
						Km/Sec	No. Lines
404.....	1944 Apr. 7	1188	7.9	+ 68	(44)	.....	.....
406.....	Apr. 9	1190	8.0	+ 70	232	−3.0	31
424.....	May 11	1222	9.3	+102	312	−3.6	47
443.....	June 4	1246	10.2	+126	388	−6.5	28
505.....	July 10–11	1282–83	11.4	+162–63	792	−5.7	21
758.....	1945 Apr. 27	1573	6.3	+ 41	142		
762.....	Apr. 28	1574	6.3	+ 42	142	−2.3	162
766.....	Apr. 29	1575	6.4	+ 43	286		
768.....	Apr. 30	1576	6.4	+ 44	12, (20)	.....	.....
769.....	Apr. 30	1576	6.4	+ 44	(10), 30	.....	.....

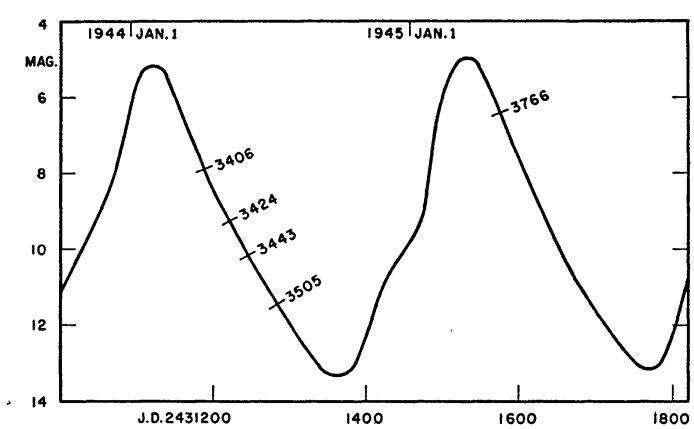

FIG. 1.—Light-curve of χ Cygni showing dates of spectrograms (see Table 2)

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ABSORPTION LINES

The spectrum as a whole with its strong *TiO* bands has the general appearance class M6, but (inconspicuous) *ZrO* bands[1] and the intensities of certain atomic lir suggest class S. The general atomic-line spectrum does not differ greatly from that R Andromedae, class Se (although the excitation temperature appears to be somewh lower), or from that of *o* Ceti, M6e. The numerous minor differences in line intensit call for future detailed study.

The mean radial velocities derived from absorption lines on spectrograms in t present series are recorded in Table 2. The velocities appear to decrease algebraica with advancing phase, but the changes are small. Likewise the differences from year year (Table 3) are small.

TABLE 3

RADIAL VELOCITIES NEAR MAXIMUM LIGHT

Observatory*	Year	Abs. Lines (Km/Sec)	Emission Lines (Km/Sec)
Potsdam	1901	+2	−20
	1902	−2	−21
Univ. Michigan	1914	.....	−17
Mount Wilson	1920	−2	−18
	1923	−3	−18
	1928	0	−14
	1938	−4	−21
	1939	+4	−12
	1944	−3	−15
	1945	−2	−15

* For references see Table 1.

The residuals yielded by various lines behave much as in R Leonis[2] and other M stars. The *K* lines λ 4044, λ 4047, are displaced shortward 3 km/sec with respect to t mean velocities adopted for absorption lines. Iron lines at excitation potential 1.0 vo are displaced longward 2 km/sec with respect to ultimate lines.

EMISSION LINES

Bright hydrogen lines are outstanding features of the maximum and early po maximum spectrum. They appear on a long series of Harvard objective-prism pho graphs, the first of which was taken in 1890. A few additional bright lines were detect on some of the Harvard plates, but it remained for slit spectrograms to record scores bright lines and to reveal the remarkable complexity of their behavior (see references Table 1). The spectrograms discussed in this paper have higher dispersion than r earlier plates and make possible a more detailed study of the lines.

The bright lines, with their remarkable changes of intensity as the phase advan from maximum toward minimum, are similar to those in R Andromedae, class Se; a in both stars the chief lines and their cyclical changes follow the general pattern p viously found to be typical of Me variables. The spectrograms of χ Cygni show emission-line spectrum richer than that previously photographed in any long-peri variable.

[1] The *ZrO* band λ 4619 is shown in P. W. Merrill, *Spectra of Long-Period Variable Stars* (Chica University of Chicago Press, 1940), Pl. II.

[2] P. W. Merrill, *Mt. W. Contr.*, No. 720; *Ap. J.*, **103**, 275, 1946.

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Lines of Fe I.—About 30 bright lines of *Fe* I were identified during the first 2 months after maximum. The number increased to almost 100 on Plate Ce 3505, phase +162 days. Most of the lines grew stronger throughout the period of observation. An exception, however, was offered by multiplets (72), (73), (74), and (75), whose lines had their maximum intensities about 110 days after maximum.

The estimated intensities of lines of multiplet (2) at various phases are in Table 4. In the last column, phase +162 days, the relative intensities are nearly the same as in the laboratory. At earlier phases the intensities are modified, probably by *TiO* band absorption, as in R Leonis.[3] The behavior of this multiplet presents another example of the general tendency of bright lines to escape from the effects of the reversing layer as the phase advances.

Intensities in multiplet (3) are approximately the same as in the laboratory, except that λ 4216 is relatively weak, probably being interfered with by λ 4215 *Sr* II. A peculiarity of multiplet (4) is the absence in emission of the leading line λ 3860. Four of the stronger lines of (5) appear to be present at the later phases but, if correctly identified, yield negative displacements of about 15 km/sec with respect to other bright lines

TABLE 4

INTENSITIES OF EMISSION LINES OF *Fe* I MULTIPLET (2)

I.A.	Lab.	χ Cygni (Phase in Days)				
		+43	+69	+102	+126	+162
4375.93	9	1	1	2.5	3	3
4427.31	10	.....	.....	1	1	3
4461.65	8	.....	.....	1	2	2.5
4482.17	4	.....	.....	.....	0.7	2
4489.74	3	.....	.....	0.5	1	1.5
4389.24	2	.....	.....	.....	.....	0.5
4435.15	2	.....	.....	.....	0.5	tr

of *Fe* I. All lines of (20), with the possible exception of λ 3820, are present in moderate or low intensity; the intensities of the first five lines relative to the others are less than in the laboratory. In (21) several lines were noted as unusually sharp; lines with upper level y^5F_2, and possibly y^5F_1, seem relatively strong.

The extraordinarily high intensities of the bright *Fe* I lines with upper level z^3G_4 have been noted in the postmaximum spectra of many long-period variables.[2, 4, 5, 6] The behavior of multiplets (24) and (42) in which these lines occur is indicated in Table 5. Very similar behavior is now clearly shown by multiplets (23) and (41), in which lines from the upper level z^5G_2 are inordinately strong (Table 6). Another possible instance, not nearly so well marked, is supplied by multiplet (21), in which lines from upper levels y^5F_1 and y^5F_2 appear relatively stronger than in the laboratory.

In χ Cygni, as in many other red variables, an outstanding bright line at 3852.6, if correctly identified with an *Fe* I line of (73), is inordinately strong, compared with other lines of the multiplet. The upper level of this line w^5D_4 produces no other lines except λ 3171, (52), in the far ultraviolet and two weak lines λ 4373, (214), and λ 5787, (625); none of these has been observed in long-period variables, and the evidence for w^5D_4 as an emphasized upper level is thus less complete than for z^3G_4 and z^5G_2.

[3] P. W. Merrill, *Pub. A.S.P.*, **58**, 304, 1946. [4] Thackeray and Merrill, *Pub. A.S.P.*, **48**, 331, 1936.

[5] Merrill and Thackeray, *Pub. A.S.P.*, **49**, 120, 1937.

[6] A. D. Thackeray, *Mt. W. Contr.*, No. 580; *Ap. J.*, **86**, 499, 1937.

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Measured displacements of 17 multiplets are arranged in Table 7 according to excita tion potential and to phase in the light-cycle. The values tend to increase algebraicall with both these parameters.[7] The change in displacement in passing from E.P. 3. (upper term) to 5.3 volts is about +8 km/sec and seems to be about the same at a phases. Similarly, the change in displacement in passing from phase +43 to +162 day about +5 km/sec, is about the same for all excitation potentials. The effect is probabl due in part to absorption components on the longward sides of many of the bright line The dark lines tend to become less intense relative to the bright lines with increasin

TABLE 5

EMISSION LINES OF *Fe* I MULTIPLETS (24) AND (42)

(24) a^5F-z^3G				(42) a^3F-z^3G			
I.A.	J	Intensity		I.A.	J	Intensity	
		Lab.	χ Cyg*			Lab.	χ Cyg*
3513.82	5–5	30	.....	4271.76	4–5	35	1
3521.26	4–4	25	8	4307.91	3–4	35	30
3526.17	3–3	15	.....	4325.76	2–3	35	1†
3466.50	5–4	3	0.3	4202.03	4–4	30	20
3483.01	4–3	3	.....	4250.79	3–3	25	0.5
3570.10	4–5	100R	.....	4147.67	4–3	10	.....
3565.38	3–4	60r	13				
3558.52	2–3	30	.....				

* Plate Ce 3505, phase +162 days. † Blend with an *Fe* I line of multiplet (2)?

TABLE 6

EMISSION LINES OF *Fe* I MULTIPLETS (23) AND (41)

(23) a^5F-z^5G				(41) a^3F-z^5G			
I.A.	J	Intensity		I.A.	J	Intensity	
		Lab.	χ Cyg*			Lab.	χ Cyg*
3581.20	5–6	250R	tr	4383.55	4–5	45r	2
3647.84	4–5	100R	.....	4404.75	3–4	30	1.5
3631.46	3–4	125R	.....	4415.12	2–3	20	1.5
3618.77	2–3	125R	tr	4294.13	4–4	15	0.5
3608.86	1–2	100R	5	4337.05	3–3	10	bl?
3589.11	5–5	8	.....	4367.91	2–2	2	1
3585.71	4–4	20	.....	4229.76	4–3	(1)	.....
3585.32	3–3	30	.....	4291.47	3–2	4	2.5
3586.98	2–2	30	4				
3528.94	5–4	.....	.....				
3540.71	4–3	3	.....				
3554.12	3–2	4	1				

* Plate Ce 3505, phase +162 days.

[7] A similar result, somewhat less marked, was shown by the *Fe* I lines in R Andromedae, as I poin out in *Mt. W. Contr.*, No. 730; *Ap. J.*, **105**, 360, Table 8, 1947.

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excitation potential and with increasing time after maximum light; naturally, as they become weaker, they have less effect in causing small displacements in the measured positions of the bright lines. The disturbing effect of the dark lines may not be the whole story; it certainly is not the main cause of the relatively large general displacement of bright lines with respect to the absorption spectrum.

Lines of Co I.—About 30 bright lines, mostly of moderate or low intensity, have been identified with *Co* I. Among the stronger are: multiplets (16) 4020.90; (28) 4131.32,

TABLE 7

RADIAL VELOCITIES FROM EMISSION LINES OF *Fe* I

(Km/Sec)

Multiplet	Upper Term		Phase (Days)					
	Desig.	E.P. (Volts)	+43	+69	+102	+126	+162	+100 (Mean)
2	z^7F	2.8	−25.6 1*	−22.5 1*	−18.0 4*	−17.0 5*	−18.8 7*	.....
3	z^7P	3.0	−23.3 2	−19.8 1	−18.8 4	−18.2 5	−19.0 5	.....
4	z^5D	3.2	.....	.....	.....	−21.5 2	−25.5 5	.....
Mean†	.....	3.1	−24.4 ..	−21.2 ..	−18.4 ..	−18.9 ..	−21.1 ..	−20.8
20	y^5D	4.1	−25.9 1	−20.6 2	−15.0 5	−18.7 10	−15.9 11	.....
21	y^5F	4.2	−20.8 2	−20.2 2	−18.6 4	−19.9 8	−18.8 10	.....
22	z^3P	4.2	.....	.....	−16.0 1	−15.2 3	−12.3 4	.....
23	z^5G	4.3	−26.1 3	−21.5 2	−16.2 3	−15.3 3	−13.9 4	.....
24	z^3G	4.4	−20.0 2	−19.1 2	−16.9 2	−17.6 3	−13.3 3	.....
Mean†	.....	4.2	−21.7 ..	−20.4 ..	−16.5 ..	−17.3 ..	−14.8 ..	−18.1
41	z^5G	4.4	.....	.....	−19.5 3	−15.0 3	−11.8 4	.....
42	z^3G	4.4	−20.8 2	−17.6 2	−15.6 3	−15.3 4	−15.5 5	.....
43	y^3F	4.6	−18.6 2	−16.6 2	−15.7 3	−16.7 4	−13.8 3	.....
45	y^3D	4.7	−13.8 1	−14.1 1	−15.6 4	−14.5 5	−13.4 5	.....
Mean†	.....	4.5	−17.7 ..	−16.1 ..	−16.6 ..	−15.4 ..	−13.6 ..	−15.9
68	x^5D	4.9	−14.8 1	.....	−15.3 1	−13.7 1	.....	.....
72	x^5P	5.3	−19.3 2	−11.4 3	− 7.8 3	− 7.1 3	− 8.6 3	.....
73	w^5D	5.4	−16.8 3	−14.4 3	−11.5 4	−11.9 5	−10.7 4	.....
74	w^5F	5.5	−13.3 4	−11.8 4	−10.1 4	−11.1 4	(−15.1) 2	.....
75	v^5D	5.5	−18.2 2	−14.0 2	−13.6 1	− 9.0 1	−10.1 1	.....
Mean†	.....	5.3	−16.5 ..	−12.9 ..	−11.7 ..	−10.6 ..	−11.1 ..	−12.6
Over-all mean	.....	4.3	−20.1 ..	−17.9 ..	−15.8 ..	−15.6 ..	−15.2 ..	−16.9

* Number of lines.

† The *lower* terms of the multiplets entering into the four sets of means have the respective excitation potentials 0.0, 0.9, 1.5, and 2.2 volts.

4118.77; (31) 3995.31; (32) 3935.96; 3997.90; (34) 3845.47; (58) 4086.30; all were measured also in the spectrum of R Andromedae. Most of the *Co* I lines appear to have their greatest intensity about 130 days after maximum light. Many of them are slightly diffuse.

Most of the lines have the lower term a^2F, E.P. 0.9 volts; three or four have the lower term b^4P, E.P. 1.9 volts. No ultimate lines have been detected; in this respect the be-

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havior differs strongly from that of lines of *Fe* I. In *Co* a general tendency toward emission is apparently overcome in ultimate lines by the absorption of the numerous atoms in the ground level; although in *Fe* this appears to be true for some of the very strong lines, at the later phases the emission of certain lines becomes so intense that it cannot be hidden by absorption as weak as that in multiplets (2) and (3), for example.

Lines of Ni I.—About 15 bright lines, mostly of low intensity, have been identified with *Ni* I. Among the stronger ones are: multiplets (30) 3783.53, 3736.81; (32) 3858.30; (52) 4331.64; all were measured also in the spectrum of R Andromedae. The *Ni* I lines reach their greatest intensities later in the cycle than do the *Co* I lines; in fact, they appear to be growing stronger throughout the observed interval after maximum up to phase +162 days.

Most of the bright *Ni* I lines have the lower term a^1D, E.P. 0.4 volts. One rather strong line, λ 4331.65 (52), has the lower term b^1D, E.P. 1.7 volts. The measured wave length of this line is persistently greater than the laboratory value by 0.1 A. This discrepancy is probably an example of the tendency of high-excitation lines to have positive displacements, although, of course, it is possible that this line is blended or even wrongly identified.

One or two strong ultimate lines are probably present in low intensity. In this respect the behavior is intermediate between that of *Co* I and that of *Fe* I. In *Co* I ultimate lines have not been observed; in *Fe* I, at the later phases, they become outstanding. Ultimate lines of *Ni* I and possibly even those of *Co* I might be strong at phases later than those observed.

Lines of Si I.—The line λ 3905 appears to consist of two components only partially separated, the one of shorter wave length being the stronger. Table 9 gives two values for this line, the upper one corresponding to the stronger component, the lower to the effective center of the whole line. For some unknown reason, the line λ 4103 has a consistent positive displacement of about 10 km/sec with respect to the velocity yielded by most other elements.

Lines of Mn I.—As in R Andromedae, the strong components of lines of the triplet near λ 4030 have a persistent negative displacement of nearly 18 km/sec with respect to the bright lines of most other elements. At the later phases a weaker component appears about 0.4 A (30 km/sec) longward. It is hoped in a future *Contribution* to make a detailed report on the behavior of these *Mn* I lines in several variables.

Lines of Zr I.—The possibility that several emission lines of *Fe* I in Me spectra are excited by the strong ultraviolet line of *Mg* II, λ 2795.52, was noted by A. D. Thackeray and P. W. Merrill[5] about ten years ago. A recent investigation by I. S. Bowen[8] shows that four lines in the postmaximum spectrum of χ Cygni (Table 13) are *Zr* I lines from the upper level, x^1G_4, probably excited by the same line of *Mg* II (see Table 12).

Line of In I.—As in several other variables, the line λ 4511 has a considerable positive residual with respect to bright lines of other elements. The accompanying tabulation which gives the apparent difference in velocity, λ 4511 minus *H*, for four stars, suggests that the presence of strong *TiO* band absorption may cause the wave length of bright λ 4511 to be measured about 0.08 A too large.

Star	*TiO*	λ 4511 − *H*
R Hya	Yes	+5.5 km/sec
R Leo	Yes	+5.8
χ Cyg	Yes	+4.8
R And	No	−1.1

[8] *Pub. A.S.P.*, **59**, 196, 1947.

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Lines of Fe II.—Numerous bright lines of *Fe* II are measurable on all the spectrograms. Most of them grow stronger with advancing phase until about +120 days, after which they tend to weaken. As in R Andromedae, multiplet (1) in the extreme ultraviolet is remarkably intense. The relative intensities of the lines are about the same as in R Andromedae,[9] except that λ 3255 is stronger, although it is still somewhat less intense than λ 3281, whereas in the laboratory it is slightly stronger. The radial velocities derived from several multiplets (Table 8) seem not to vary with phase or with excitation potential.

Forbidden lines begin to appear about 100 days after maximum and strengthen rapidly during the next 2 months. Their velocities agree with those from the permitted lines.

Lines of Ti II.—A number of lines of *Ti* II persist in low intensity during most of the observed portion of the light-cycle, the phase of greatest intensity being perhaps +120 days, with a noticeable weakening by +160 days.

TABLE 8

RADIAL VELOCITIES FROM EMISSION LINES OF *Fe* II

Multiplet	Upper Term		Phase (Days)					
	Desig.	E.P. (Volts)	+43	+69	+102	+126	+162	+100 (Mean)
1	z^6D	4.8	−15.2 5*	.....	−20.3 3*	.....	−15.1 7*	−16.2
3	z^6D	4.8	−11.4 1	−10.9 1*	−11.1 1	−12.6 1*	−12.3 1	−11.7
6	z^4D	5.5	.....	.....	.....	.....	−17.2 1	−17.2
27	z^4D	5.5	−15.3 3	−15.2 2	−13.3 4	−14.4 4	−14.6 5	−14.4
28	z^4F	5.6	(− 7.5) 1	(− 7.8) 1	− 9.4 1	−13.7 3	− 7.8 3	− 9.9
37	z^4F	5.5	.....	.....	.....	−11.8 2	−10.7 1	−11.4
38	z^4D	5.5	−13.1 1	.....	−13.8 2	−15.1 3	.....	−14.3
42	z^6P	5.3	−15.2 1	−16.2 1	−15.1 1	−16.9 1	.....	−15.8
Weighted mean	.....	.....	−14.7	−14.4	−15.3	−14.2	−14.6	−14.6

* Number of lines.

General.—The velocities yielded by the bright lines of several elements are compared in Table 9. As in R Andromedae, the displacements of *H*, *Fe* II, *Ti* II, and *Mg* I in multiplet (3) agree with each other and vary little with phase. Moreover, *Mg* I λ 4571 and the lines of *Fe* I, which at first appear as bright fringes on the shortward sides of dark lines and thus have negative residuals, tend as the cycle advances to assume the standard velocity.

Displacements of bright lines of various elements (e.g., *H* and *Fe* II) exhibit smaller differences in R Andromedae and χ Cygni than they do in Me variables; and variations with phase also are smaller. At the same time the light-ranges of R Andromedae and χ Cygni are relatively large. It thus appears doubtful that the complex behavior of the bright lines will find a simple explanation in terms of volume pulsation.

Displaced lines.—The general over-all displacement of the bright lines shortward with respect to the absorption spectrum has been observed and discussed for many years. Not so well known are the additional shortward displacements exhibited by bright lines which develop usually at the later phases in certain very strong lines of *Ca* II, *Al* I, *Cr* I, *Fe* I, and one or two other elements. These lines are slightly wide and often appear

[9] P. W. Merrill, *Mt. W. Contr.*, No. 730; *Ap. J.*, **105**, 360, Table 9, 1947.

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slightly unsymmetrical, being more sharply bounded on the longward side. Their displacements, greatest for the lines which are strongest in absorption, tend to decrease with phase.

The displaced lines of *Ca* II, previously observed in the spectra of several long-period variables, become especially intense in χ Cygni. The anomalous reversal of the normal intensities of H and K is marked (Table 10). As in other variables, the intensities increase and the negative displacements decrease with advancing phase.

The strong *Al* I lines, λ 3944 and λ 3961 between H and K, also exhibit displaced emission (Table 10), although the intensities are much less and are more nearly the same in both lines than in the *Ca* II pair.

TABLE 9

RADIAL VELOCITIES FROM EMISSION LINES

(Km/Sec)

Element	Line or Multiplet	Phase (Days)					
		+43	+69	+102	+126	+162	+100 Mean
H	Many	−15.0	−15.6	−13.5	−14.5	−13.6	−14.4
Fe I	Table 5	−20.1	−17.9	−15.8	−15.6	−15.2	−16.9
Mg I	λ 4571	−21.8	−16.2	−15.7	−16.1	.....	.....
Mg I	(3)	−17.0	−17.6	−16.7	−17.8	−16.2	−17.1
Si I	λ 3905	−20.1	−21.1	−22.2	−22.1	.....	−21.4
		−13.4	−13.4	−12.2	−12.1	−10.9	−12.4
Si I	λ 4103	− 6.5	− 6.1	− 7.0	− 5.7	− 5.0	− 6.0
Mn I	(2)	−34.	−39.	−33.	−32.	−29.	−33.
In I	λ 4511	−11.7	−10.0	− 8.8	− 9.1	.....	− 9.9
Fe II	Table 6	−14.7	−14.4	−15.3	−14.2	−14.6	−14.6
[*Fe* II]	Several*	.....	.....	−14.8	−14.4	−13.5	−14.0
Ti II	(13)	−15.2	−13.8	−12.9	−14.3	−14.7	−14.2
Abs. lines	.....	− 2.3	− 3.0	− 3.6	− 6.5	− 5.7	− 4.2

* At the three epochs the numbers of lines measured were 3, 9, and 14, respectively.

TABLE 10

DISPLACEMENTS OF BROAD EMISSION LINES

(Corrected for Absorption-Line Velocities, Table 2)

Phase (Days)	Ca II				Al I			
	λ 3933		λ 3968		λ 3944		λ 3961	
	Int.	Displ. (Km/Sec)	Int.	Displ (Km/Sec)	Int.	Displ. (Km/Sec)	Int.	Displ. (Km/Sec)
+ 43	.....	.....	1	−100	tr	−64	tr	−72
+ 69	.....	.....	1	− 96	tr	.....	tr	.....
+102	3	−102	6	− 82	1	−58	1	−59
+126	4	− 94	7	− 67	1	−42	1	−45
+162	5	− 67	9	− 53	1	−29	1	−29

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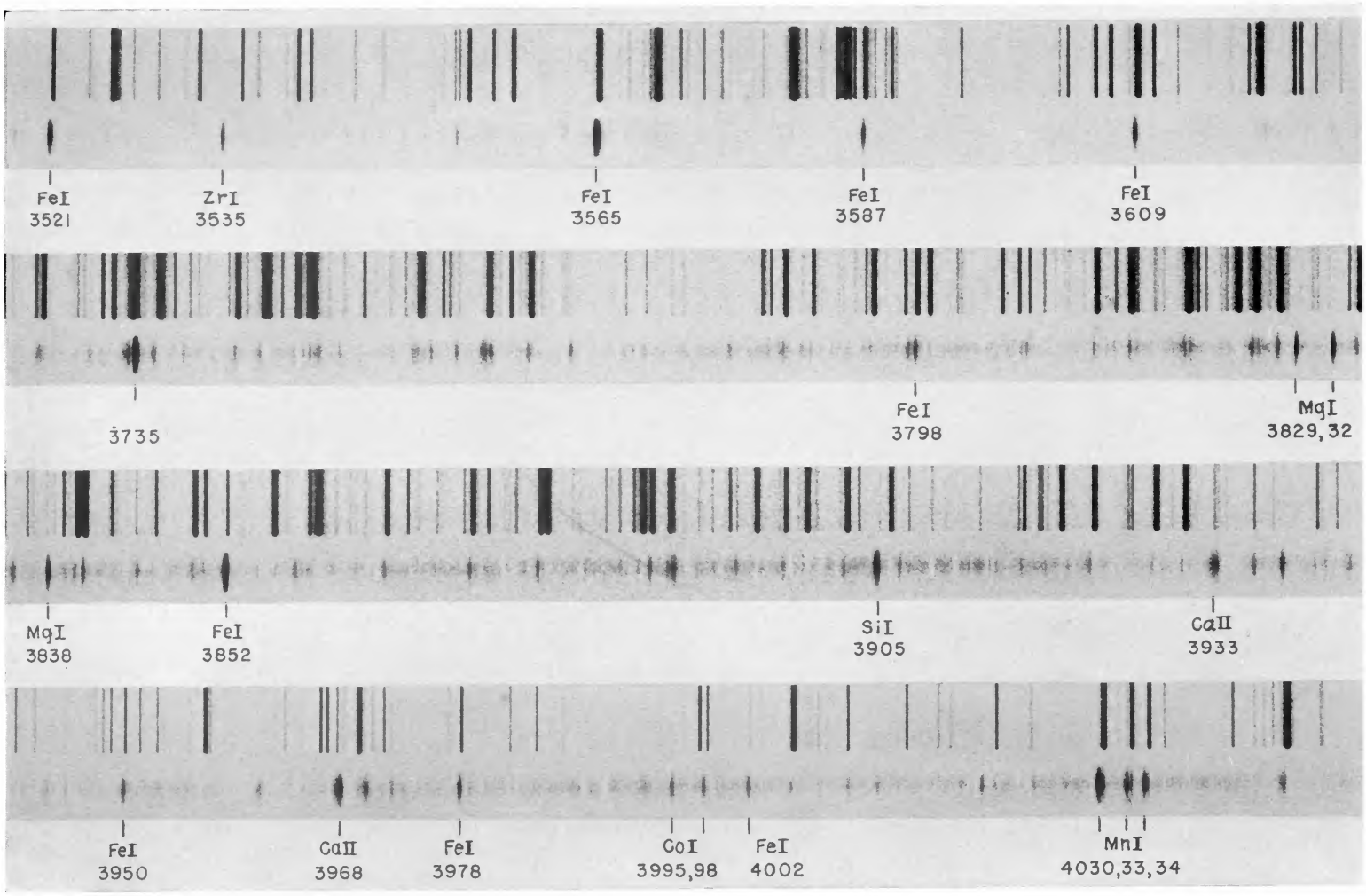

FIG. 2.—Ultraviolet spectrum of χ Cygni, 162 days after maximum light. Plate Ce 3505, 1944, July 10–11

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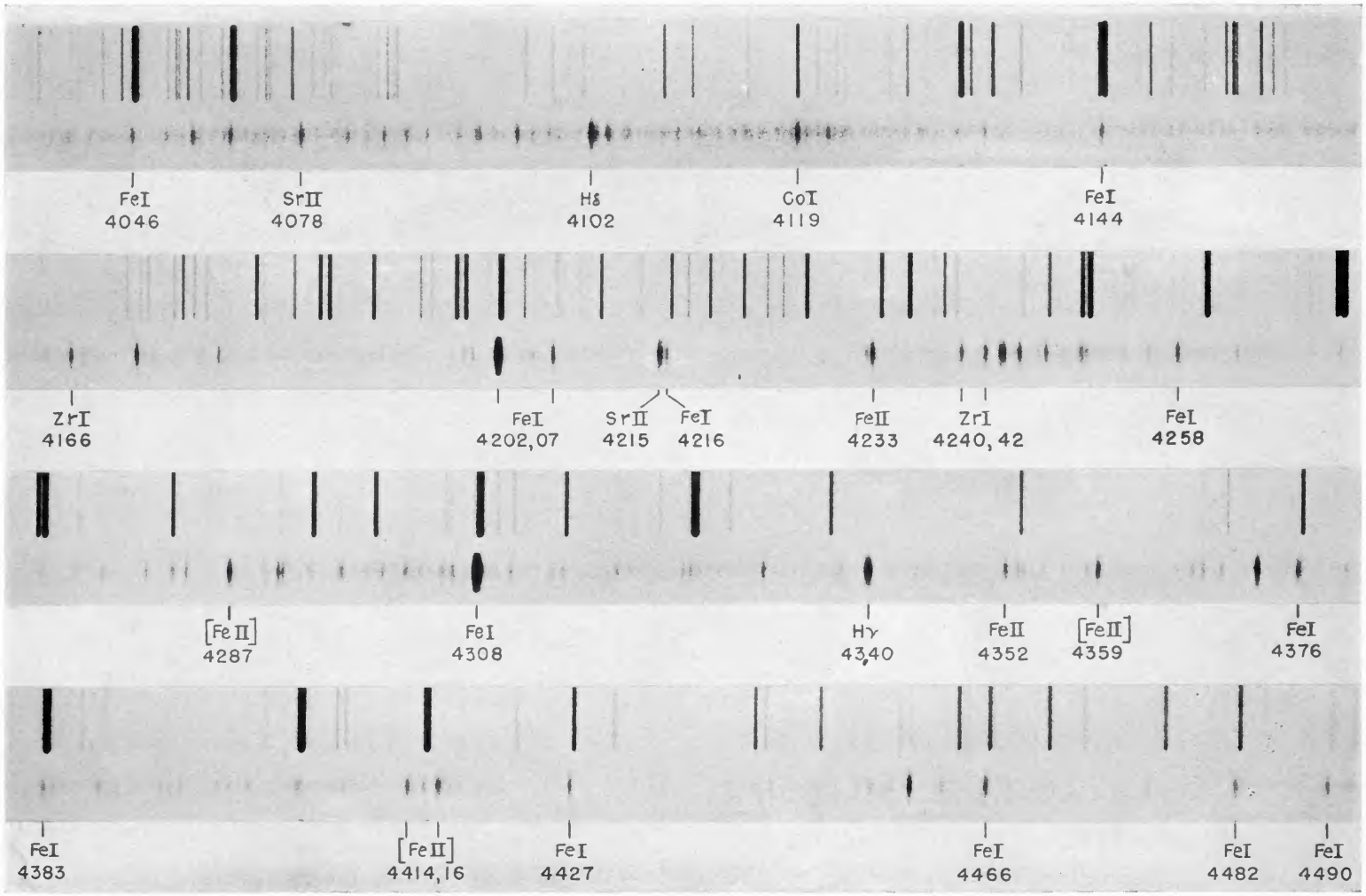

FIG. 3.—Spectrum of χ Cygni λλ 4050–4500, 162 days after maximum light. Plate Ce 3505, 1944, July 10–11

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The *Sr* II lines, λ 4077 and λ 4215, also exhibit shortward displacements which decrease with advancing phase. The line λ 4215 is the stronger of the two (although weaker in the laboratory) and has somewhat smaller shortward displacements. The behavior thus has a resemblance to that of the analogous *Ca* II lines, λ 3933 and λ 3968.

An ingenious method of determining the probable structure of the strong unobserved lines λ 2795 and λ 2802 of *Mg* II, suggested by Ira S. Bowen, is outlined in Table 11,

TABLE 11

FLUORESCENT EXCITATION BY ULTRAVIOLET LINES OF *Mg* II

Exciting Line	Other Close Lines		
	Element	Δλ	Observed Fluorescence
Mg II 2795.52	*Co* I	+0.71 A	No
	Co I	+0.30	No
	Fe I	+0.02	No
	Zr I	−0.39	Yes
	Fe I	−0.52	Yes
	Fe I	−0.83	Yes
Mg II 2802.70	*Fe* I	+0.47	No
	Ti I	−0.23	Yes?

TABLE 12

ATOMIC LEVELS YIELDING ABNORMALLY INTENSE EMISSION LINES

Element	Observed Lines	Upper Level	Line from Low Term	Exciting Line
Fe I	Table 5	z^3G_4	2795.01	*Mg* II 2795.53
Fe I	Table 6	z^5G_2	.....	.....
Fe I	3852.57	w^5D_4	2794.70	*Mg* II 2795.53
Zr I*	3535.16, 4166.51, 4240.44, 4242.61	x^1G_4	2795.14	*Mg* II 2795.53
In I	4511.31	$6^2S_{1/2}$	4101.76	*Hδ* 4101.74
Sc I	3907.48	$y^2P_{2\frac{1}{2}}$	3933.38	*Ca* II 3933.66
Ti I	4372.38	v^1P_1	2802.47	*Mg* II 2802.70
	4400.60			
Ga I	4172.05	$5^2S_{1/2}$	4032.98	*Mn* I 4033.07

* I. S. Bowen, *Pub. A.S.P.*, **59**, 196, 1947.

which lists lines of other elements lying near the *Mg* II lines. It appears that only those lines of somewhat shorter wave lengths than the *Mg* II lines produce fluorescent effects. This result suggests that the emission portions of the *Mg* II lines are displaced shortward as in the H and K lines of *Ca* II.

Atomic levels that yield abnormally intense emission lines, together with the probable exciting lines, are collected in Table 12. The identification of the observed lines ascribed to *Sc* I, *Ti* I, and *Ga* I is not considered wholly certain.

The bright lines measured on Plate Ce 3505, phase +162 days, magnitude 11.4, are listed in Table 13. The measured wave lengths have been corrected for a velocity (reduced to the sun) of −14.4 km/sec. Figures 2 and 3 show how strong many of these lines

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TABLE 13

BRIGHT LINES MEASURED AT PHASE +162 DAYS (PLATE Ce 3505)

I.A.	Int.	Change with Phase*	Ident.	I.A.	Int.	Change with Phase*	Ident.
3227.70....	tr	.....	.73 *Fe* II (6)	3761.32....	1	+110	.32 *Ti* II (13)
55.90....	3	.....	.88 *Fe* II (1)	63.44....	1	.....	.....
77.34....	8	.....	.35 *Fe* II (1)	63.78....	1	.....	.79 *Fe* I (21)
81.29....	4	.....	.29 *Fe* II (1)	64.14....	1.5	.....	.21 *Fe* I (74)?
95.80....	2	.....	.81 *Fe* II (1)	67.18....	1	.....	.19 *Fe* I (21)
3302.81....	1	.....	.86 *Fe* II (1)	70.4.....	1.5	Dec.	.63 *H* 11
03.46....	1	.....	.47 *Fe* II (1)	74.86....	tr	+120	.82 *Fe* I (73)
13.99....	tr	.....	.00 *Fe* II (1)	75.57....	tr	+130	.57 *Ni* I (33)
71.9.....	tr	.....	.99 *Ni* I (7)	76.51....	tr	+110	.45 *Fe* I (74)
3439.0.....	tr	.....	.98 *Mn* II (1)	79.62....	0.5	Dec.	.49 *Fe* I (74)
44.33....	tr	.....	.31 *Ti* II (6)	83.47....	1	.....	.53 *Ni* I (30)
60.1.....	tr	.....	.04 *Mn* II (1)	86.6.....	tr	Dec.	.68 *Fe* I (22)
61.5.....	tr	.....	.50 *Ti* II (6)	87.53....	tr	Dec.	.....
66.53....	0.5	Inc.	.51 *Fe* I (24)	87.87....	0.5	Inc.?	.88 *Fe* I (21)
77.2.....	tr	.....	.18 *Ti* II (6)	89.36....	tr n	.....	.....
3518.4.....	0.5	Inc.?	.34 *Co* I (36)?	90.14....	tr	.....	.10 *Fe* I (22)
21.25....	8	Inc.!	.26 *Fe* I (24)	90.96....	tr	.....	.....
30.9.....	tr	Dec.	.....	94.92....	tr	.....	.00 *Fe* I (21)
35.22....	4	Inc.	.16 *Zr* I (59)	97.98....	1 n	Dec.	.90 *H* 10
45.3.....	tr	Dec.	.19 *V* II (5)	98.53....	2	Inc.	.51 *Fe* I (21)
49.95....	0.5	Dec.	.87 *Fe* I (48)	99.65....	0.5 n	.....	.55 *Fe* I (21)
51.54....	tr	.....	.53 *Ni* I (5)?	3807.08....	tr	.....	.14 *Ni* I (33)
54.14....	1	.....	.12 *Fe* I (23)	07.50....	0.5	Dec.	.53 *Fe* I (73)
65.41....	14	Inc.!	.38 *Fe* I (24)	12.98....	1	.....	.96 *Fe* I (22)
81.2.....	tr	.....	.19 *Fe* I (23)	13.99....	tr	Dec.	.....
86.98....	4	Inc.	.98 *Fe* I (23)	14.61....	0.5	.....	.58 *Ti* II (12)
3608.85....	5	Inc.	.86 *Fe* I (23)	15.86....	1	Inc.	.84 *Fe* I (45)
18.6.....	tr	.....	.76 *Fe* I (23)?	16.44....	tr	+110	.34 *Fe* I (73)?
19.3.....	0.5	.....	.39 *Ni* I (35)	20.33....	1	Inc.	.43 *Fe* I (20)
24.7.....	tr	.....	.73 *Ni* I (2)	20.79....	0.5	+130	.....
79.6.....	tr	.....	.92 *Fe* I (5)	25.76....	1.5	Inc.	.88 *Fe* I (20)
85.21....	tr	.....	.19 *Ti* II (14)	26.25....	0.5	+130	.....
3705.4.....	tr	.....	.57 *Fe* I (5)	27.83....	1.5	Inc.	.82 *Fe* I (45)
09.1.....	tr n	Dec.	.25 *Fe* I (21)	29.33....	4	.....	.35 *Mg* I (3)
11.3.....	tr	.....	.30 *Fe* I (75)?	32.27....	4	.....	.30 *Mg* I (3)
11.99....	tr	Dec.	.97 *H* 15	34.16....	1	Inc	.22 *Fe* I (20)
19.52....	3 n	Inc.	.94 *Fe* I (5)	35.42....	1.5 n	Dec	.39 *Hη*
21.33....	tr	Dec.	.28 *Fe* I (75)	38.27....	5	.....	.29 *Mg* I (3)
21.93....	tr	Dec.	.94 *H* 14	40.39....	0.5	Inc.	.43 *Fe* I (20)
27.43....	2 n	.....	.62 *Fe* I (21)	41.08....	2	Inc.	.05 *Fe* I (45)
31.62....	1 n	.....	.03 *Cr* I (2)?	45.52....	1	.....	.47 *Co* I (34)
34.44....	0.5 n	.....	.37 *H* 13	49.95....	0.5	.....	.97 *Fe* I (20)
34.76....	2	.....	.87 *Fe* I (21)	50.78....	tr	.....	.82 *Fe* I (22)
35.31....	10	Inc.!	.....	52.66....	10	.....	.57 *Fe* I (73)
36.84....	2 n	Inc.	.13 *Fe* I (5)	58.28....	1	Inc.?	.30 *Ni* I (32)
39.21....	1	Inc.	.22 *Ni* I (2)?	61.23....	tr n	+130	.16 *Co* I (33)
41.6.....	tr	.....	.63 *Ti* II (72)	65.52....	1	.....	.52 *Fe* I (20)
43.33....	0.5	.....	.36 *Fe* I (21)	72.49....	1	.....	.50 *Fe* I (20)
45.34....	0.5 n	.....	.56 *Fe* I (5)	78.02....	1.5	.....	.02 *Fe* I (20)
48.07....	0.5	.....	.26 *Fe* I (5)	78.41....	tr	.....	.57 *Fe* I (4)?
48.89....	0.5	Dec.	.04 *Ni* I (1)?	84.70....	tr n	+120	.60 *Co* I (32)
49.38....	1	.....	.49 *Fe* I (21)	86.11....	tr	Inc.?	.28 *Fe* I (4)
50.03....	1 n	Dec.	.15 *H* 12	87.08....	1.5	Inc.	.05 *Fe* I (20)
51.79....	0.5	.....	.....	88.54....	1	.....	.52 *Fe* I (45)
57.98....	1	.....	.24 *Fe* I (21)	89.02....	2 n	Dec.	.05 *Hζ*
3759.29....	1	+110	.29 *Ti* II (13)	3894.14....	0.5	Dec.	.07 *Co* I (34)

* Inc., increasing intensity; Inc.!, marked increase from phase +126 days to +162 days; Dec., decreasing intensity. For lines which appear to have a maximum intensity between phases +43 days and +162 days, the approximate phase of maximum is indicated.

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TABLE 13—*Continued*

I.A.	Int.	Change with Phase*	Ident.	I.A.	Int.	Change with Phase*	Ident.
3898.01....	0.5	.....	.01 *Fe* I (20)	4132.0.....	tr	Dec.	.06 *Fe* I (43)
3903.00....	0.5	.....	.95 *Fe* I (45)	34.29....	0.5	Inc.	.34 *Fe* I (3)
05.58....	6	.....	.53 *Si* I (3)	43.88....	2	Inc.	.87 *Fe* I (43)
07.41....	0.5	.....	.48 *Sc* I (8)?	56.8.....	tr n	Dec.	.....
12.97....	tr	.....	.97 *Ni* I (15)	64.68....	tr	Dec.	.64 *Ni* I (28)
14.53....	tr	+130	.48 *Fe* II (3)?	66.58....	3	+150	.51 *Zr* I
17.21....	0.5	.....	.18 *Fe* I (20)	67.53....	tr	+120	.....
22.75....	1	Inc.?	.91 *Fe* I (4)	72.05....	5	Inc.!	.05 *Ga* I (1)?
27.81....	0.5	.....	.92 *Fe* I (4)	72.76....	0.5	Inc.?	.75 *Fe* I (19)
30.17....	0.5	.....	.30 *Fe* I (4)	73.48....	1	+120	.45 *Fe* II (27)
32.90....	4 n	Inc.	.66 *Ca* II (1)	77.65....	0.5	.....	.....
36.02....	1 n	+130	.96 *Co* I (32)	78.89....	2	+130	.86 *Fe* II (28)
38.32....	3	+110	.29 *Fe* II (3)	99.95....	1	Inc.	.97 *Fe* I (3)
40.94....	0.5	.....	.88 *Fe* I (20)	4202.07....	20	Inc.	.03 *Fe* I (42)
41.7.....	0.5 n	.....	.73 *Co* I (17)	04.2.....	0.5	+110	.....
43.74....	1 n	.....	.01 *Al* I (1)	06.63....	3	Inc.	.70 *Fe* I (3)
45.27....	0.5	+130	.21 *Fe* II (3)?	15.51....	5	Inc.	.52 *Sr* II (1)
50.03....	3	.....	.95 *Fe* I (72)	16.05....	2	Inc.	.18 *Fe* I (3)
61.25....	1 n	.....	.52 *Al* I (1)	32.67....	1	Inc.	.72 *Fe* I (3)
67.89....	8 n	Inc.	.47 *Ca* II (1)	33.15....	4	.....	.17 *Fe* II (27)
70.06....	2 n	Dec.	.07 *Hε*	40.48....	2	+140	.44 *Zr* I
72.21....	1 n	Inc.	.17 *Ni* I (29)	42.49....	2	Inc.!	.61 *Zr* I, —
77.81....	3	.....	.74 *Fe* I (72)	43.68....	3	Inc.!	.....
78.7.....	tr	.....	.65 *Co* I (17)	44.00....	3	Inc.	.98 *Fe* II (21F)
93.0.....	0.5 n	.....	.....	45.26....	3	Inc.!	.....
95.35....	2	.....	.31 *Co* I (31)	46.83....	tr	.....	.83 *Sc* II (7)
97.97....	2	+120	.90 *Co* I (32)	47.45....	2	Inc.!	.....
4001.75....	2	+110	.67 *Fe* I (72)	50.31....	1	Inc.!	.....
05.28....	1	Inc.	.25 *Fe* I (43)	50.73....	0.5	.....	.79 *Fe* I (42)
09.94....	0.5	+110	.71 *Fe* I (72)?	53.9.....	tr n	.....	.35 *Cr* I (1)
12.38....	0.5	+120	.37 *Ti* II (11)	55.88....	0.5	Inc.?	.....
20.92....	1	.....	.90 *Co* I (16)	58.28....	3	Inc.	.32 *Fe* I (3)
25.09....	tr	+120	.14 *Ti* II (11)	63.20....	1	Inc.	.....
30.52....	6	+130	.76 *Mn* I (2)	71.76....	1	Inc.	.76 *Fe* I (42)
30.91....	3	Inc.!		74.6.....	0.5	Inc.?	.80 *Cr* I (1)
				76.85....	1	Inc.!	.83 *Fe* II (21F)
32.88....	3	Inc.	.07 *Mn* I (2)	80.47....	1.5	Inc.!	.....
33.26....	2	Inc.!		82.65....	2 n	Inc.	.41 *Fe* I (71) bl?
				83.83....	2	Inc.!	.....
34.29....	3	+160	.49 *Mn* I (2)	87.41....	5	Inc.!	.40 *Fe* II (7F)
34.59....	2	Inc.!		90.27....	tr	.....	.22 *Ti* II (41)
45.68....	2 n	.....	.80 *Fe* I (43)	91.44....	3	Inc.	.47 *Fe* I (3)
63.59....	1.5	.....	.60 *Fe* I (43)	91.85....	2	Inc.!	.....
66.42....	0.5	.....	.36 *Co* I (36)	94.2.....	0.5 n	.....	.13 *Fe* I (41)
68.60....	2	.....	.62 *S* II (1F)?	96.69....	2 n	Inc.	.57 *Fe* II (28)?
71.26....	0.5	Inc.	.....	4300.10....	0.5	.....	.05 *Ti* II (41)
71.75....	2	.....	.74 *Fe* I (43)	02.01....	1 n	+130	.93 *Ti* II (41)
77.68....	2	Inc.	.71 *Sr* II (1)	03.16....	0.5	+130	.17 *Fe* II (27)
81.46....	0.5	Inc.!	.....	05.89....	1.5	Inc.	.90 *Fe* II (21F)
84.49....	1	Inc.!	.....	07.92....	30	Inc.	.91 *Fe* I (42)
86.34....	1	+120	.30 *Co* I (58)	09.7.....	tr	+130	.62 *Y* II (5)
88.01....	1	Inc.!	.....	14.23....	tr n	.....	.....
90.7.....	0.5 n	Inc.?	.....	15.07....	0.5 n	Inc.?	.98 *Ti* II (41)
92.30....	2 n	Inc.?	.39 *Co* I (29)	19.61....	1 n	Inc.!	.62 *Fe* II (21F)
95.54....	2	Inc.	.....	20.6.....	tr n	Inc.?	.....
4101.7.....	6 n	Dec.	.74 *Hδ*	25.09....	0.5	Dec.	.....
03.06....	2 n	Dec.	.93 *Si* I (2)	25.68....	0.5	+150	.76 *Fe* I (42)
08.84....	0.5	Inc.	.....	31.77....	2	Inc.	.....
18.81....	3	+140	.77 *Co* I (28)	38.14....	0.5 n	Inc.	.....
19.53....	tr	Dec.	.....	40.44....	8 n	Dec.	.47 *Hγ*
4121.36....	1	+140	.32 *Co* I (28)	4347.0.....	tr n	.....	See note

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TABLE 13—*Continued*

I.A.	Int.	Change with Phase*	Ident.
4351.78....	2	.....	.76 *Fe* II (27)
52.78....	0.8	Inc.	.78 *Fe* II (21F)
53.25....	0.5	Inc.	.....
54.97....	tr	Inc.	.....
56.60....	tr	Inc.?	.....
58.43....	0.8	Inc.	.37 *Fe* II (21F)
59.35....	3	Inc.!	.34 *Fe* II (7F)
67.13....	0.5	Inc.!	.....
67.99....	1	+120	.90 *Fe* I (41)
69.53....	1	Inc.!	.....
72.50....	6	+140	.38 *Ti* I (277)?
75.88....	3	+140	.93 *Fe* I (2)
79.81....	1	Inc.	.....
83.52....	2 n	+120	.54 *Fe* I (41)
84.40....	0.5 n	Inc.	.....
85.36....	tr	.....	.38 *Fe* II (27)
88.15....	1 n	Inc.!	.....
89.23....	tr	Inc.?	.24 *Fe* I (2)
93.35....	0.8	Inc.!	.....
94.07....	tr	+120	.12 *Fe* I (41)
95.04....	0.5	+120	.03 *Ti* II (19)
98.02....	tr	.....	.02 *Y* II (5)
99.02....	0.5	Inc.?	.....
4400.64....	1	Inc.	.....
04.81....	1 n	+120	.75 *Fe* I (41)
08.5.....	tr	.....	.42 *Fe* I (68)?
13.81....	2.5 n	Inc.!	.78 *Fe* II (7F)
15.16....	1.5 n	+120	.12 *Fe* I (41)
16.32....	3.5	Inc.!	.27 *Fe* II (6F)
27.25....	3	Inc.	.31 *Fe* I (2)
4435.0.....	tr	+120?	.15 *Fe* I (2)?
4442.4.....	tr	+120?	.34 *Fe* I (68)
43.9.....	0.5	+120	.80 *Ti* II (19)?
52.13....	1 n	Inc.	.11 *Fe* II (7F)
55.34....	4	Inc.!	.....
57.94....	1 n	Inc.!	.95 *Fe* II (6F)
61.60....	3	Inc.	.65 *Fe* I (2)
66.58....	tr	Inc.?	.57 *Fe* I (2)
74.89....	tr	Inc.	.91 *Fe* II (7F)
82.09....	2	Inc.!	.17 *Fe* I (2)
89.65....	2	Inc.	.74 *Fe* I (2)
94.56....	tr	Inc.?	.57 *Fe* I (68)
4501.26....	0.5	+120?	.27 *Ti* II (31)
Phase +126 Days (Ce 3443)			
4511.39....	2	Dec.	.31 *In* I (1)
18.72....	1	.....	.....
20.20....	1	.....	.22 *Fe* II (37)
22.64....	0.5	.....	.63 *Fe* II (38)
28.63....	2 n	.....	.61 *Fe* I (68)
31.20....	0.5	.....	.15 *Fe* I (39)
34.05....	1 n	.....	See note
49.48....	2	.....	.47 *Fe* II (38)
55.97....	1 n	.....	.89 *Fe* II (37)
71.08....	15	Inc.	.10 *Mg* I (1)
83.79....	2	.....	.83 *Fe* II (38)
4861.18....	10 n	.....	.33 *Hβ*
4923.89....	2	.....	.92 *Fe* II (42)

NOTES TO TABLE 13

λ 3565.41 The line appears to consist of two components, one at λ 3565.34, intensity 10; another at λ 3565.60, intensity 3. The longward component developed rapidly after phase +100 days.

λ 3757.98 The line appears to consist of two components, one at λ 3757.84, intensity 1; the other at λ 3758.19, intensity 0.5.

λ 3852.66 At earlier phases a faint companion lies 0.4 A shortward.

λ 3905.58 At earlier phases components were measured at λ 3905.43 and λ 3905.71. The intensity on the longward component gradually increased from 0.4 to about 1.4 times the intensity of the other.

λ 3945.27 Possibly blended with λ 3945.33 *Co* I (29).

λ 4045.68 At earlier phases this line may have a component (possibly a companion line of another element) about 0.3 A shortward.

λ 4092.30 Possibly double, with components of nearly equal intensity at λ 4092.12 and λ 4092.44.

λ 4215.51 Unsymmetrical; main component at λ 4215.43, with a fringe or faint component at λ 4215.62.

λ 4247.45 The remarkable preminimum bright line previously observed with low dispersion (P. W. Merrill, *Mt. W. Contr.*, No. 539; *Ap. J.*, **83**, 272, 1936) at λ 4245 is here resolved into several lines from λ 4242.49 to λ 4247.45. The line λ 4242.49 is probably a blend of λ 4242.61 *Zr* I with an unidentified line, whose wave length might be λ 4242.4.

λ 4315.07 Possibly blend with λ 4315.09 *Fe* I (71).

λ 4325.68 Possibly blend with λ 4325.74 *Fe* I (2).

λ 4347.0 Possibly blend of λ 4346.85 *Fe* II (21F) with λ 4347.24 *Fe* I (2).

λ 4352.78 Possibly blend with λ 4352.74 *Fe* I (71).

λ 4375.88 At the earlier phases the wave length was shorter by about 0.1 A. The increasing relative intensity of emission with respect to absorption on the longward side may account for the slight increase in wave length.

λ 4511.39 This line and the following ones were measured at phase +126 days, Plate Ce 3443. Wave lengths greater than λ 4510 are not on Plate Ce 3505.

λ 4534.05 Possibly a blend of λ 4533.97 *Ti* II (50) and λ 4534.17 *Fe* II (37).

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are, in comparison with the underexposed background of continuous spectrum. Unidentified bright lines from Table 13 and a few measured only at earlier phases are collected in Table 14; the wave lengths, derived from all available measures, may differ slightly from those in Table 13.

In Tables 13 and 14 an attempt has been made to give an idea of the behavior of each line as the cycle advances. The indicated phase of maximum observed intensity is, of course, only approximate but should be helpful in checking possible identifications. The lines may be grouped according to the time of maximum intensity as follows:

I. Lines whose intensities decrease steadily after maximum light. The phase of greatest intensity in the present series of spectrograms is +40 days. Two of the strongest unidentified lines are λ 4138.53 and λ 4521.72.

TABLE 14

SURVEY OF UNIDENTIFIED BRIGHT LINES

I.A.	Max. Obs. Int.		I.A.	Max. Obs. Int.		I.A.	Max. Obs. Int.	
	Int.	Phase* (Days)		Int.	Phase* (Days)		Int.	Phase* (Days)
3421.2.....	tr	+(80)	3952.86.....	1	+ 40	4283.83.....	2	+ 160!
3531.0.....	0.5	+(60)	92.96.....	0.5	+ 150	91.85.....	2	+ 160!
3724.4.....	0.5	+ 90	4081.46.....	0.5	+ 160!	92.45.....	1	+ 70
26.9.....	0.5	+ 70	84.49.....	1	+ 160!	4325.11.....	1	+ 70
35.34.....	10	+ 160!	88.01.....	1	+ 160!	31.75.....	2	+ 160
42.31.....	0.5	+ 40	95.52.....	2	+ 140	38.14.....	0.5 n	+ 160
51.81.....	0.5	+ 140	98.95.....	1	+ 50	53.25.....	0.5	+ 160
63.3–63.4..	2	+ 110	4119.46.....	1	+ 40	67.13.....	0.5	+ 160!
85.99.....	0.5	+ 40	34.88.....	0.5	+ 110	69.53.....	1	+ 160!
87.54.....	1	+ 40	38.53.....	2	+ 40	72.52 *Ti* I?	6	+ 140
94.52.....	1	+ 40	56.86.....	1	+ 70	79.81.....	1	+ 160
3814.01.....	2	+ 100	72.05 *Ga* I?	5	+ 160!	84.40.....	0.5 n	+ 160
20.8.....	1	+ 130	4243.68.....	3	+ 160!	88.15.....	1 n	+ 160!
22.87.....	0.5	+ 40	45.25.....	3	+ 160!	93.35.....	0.8	+ 160!
26.3.....	1	+ 130	47.45.....	2	+ 160!	4400.6.....	1	+ 160
44.92.....	1	+ 40	50.31.....	1	+ 160!	55.34.....	4	+ 160!
3907.44 *Sc* I?.	1	+ 120	55.90.....	0.5	+ 160	4518.72.....	1	+(160)
13.6–13.7..	1	+ 100	63.19.....	1	+ 160	21.72.....	3	+ 40
52.2.....	1	+ 40	80.47.....	1.5	+ 160!			

* An exclamation mark (160!) indicates that the line was much stronger than at earlier phases.

II. Lines which have their greatest intensities at intermediate phases. Lines of *Co* I, *Fe* II, *Ti* II, *Y* II, and high excitation lines of *Fe* I fall in this group; other lines of *Fe* I appear to be intermediate between groups II and III.

III. Lines which grow stronger rapidly after phase +130 days. These are marked "Inc.!" in Table 13 or "+160!" in Table 14. Most of the identified lines are due to [*Fe* II]. Whether most of the unidentified lines of type III also are forbidden remains to be determined.

The numerous unidentified lines constitute an engaging problem. The same lines probably occur in many or all typical variables of classes Me and Se, although they appear to be especially strong in χ Cygni. Further data concerning these lines (and possibly additional lines) can eventually be obtained from suitable spectrograms taken closer to minimum light.

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